SiteProof

Implementation & Domain Reference

A South African construction evidence, claims and compliance platform

Prepared for: Lisan Al-Gaib, Self Hawk Tech (Pty) Ltd **Date:** 1 September 2026 **Status:** Working reference document, not a specification to build from unverified

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0. How to use this document

This document does three separate jobs, and you should read it in three separate passes.

Part 1 is the strategic frame — what the product is, who buys it, what it competes with, and how it should be priced in South Africa. Read this first and argue with it.

Part 2 is the construction domain. This is the part you cannot skip. You are a software developer entering an industry with two hundred years of contractual convention, a dense statutory overlay, and a vocabulary that will expose you instantly if you get it wrong. Part 2 is written to be read like a textbook chapter, and it is deliberately longer than the engineering sections, because the engineering is the part you already know how to do.

Part 3 and 4 are system design and stack. These assume Part 2 is understood.

Part 5 is a resource library — every link in this document, organised by topic, so you can go deeper on any single area.

A word on verification

Every legal, contractual and regulatory statement in this document was checked against a source at the time of writing and those sources are linked inline and in Part 5. That is not the same as being correct today.

Three specific decay risks:

1. **Contract editions renumber.** Clause references given here are for JBCC Principal Building Agreement Edition 6.2 (May 2018). If a newer edition is current, the clause numbers will have moved. Never hard-code a clause reference into generated legal correspondence without a version-aware lookup table.
2. **Regulatory thresholds are amended by notice.** The Construction Regulations permit threshold has already been changed once by exemption notice. Check the current position with the Department of Employment and Labour, not with a blog.
3. **This document is not legal advice and neither are you.** Before you ship a feature that tells a contractor "your notice deadline has passed," have a construction attorney or an experienced quantity surveyor review the logic. Getting this wrong causes real financial harm to your customer and creates real liability for you.

The working assumption throughout: **verify at source before writing code that depends on it.**

PART 1 — STRATEGIC FRAME

1.1 What the product actually is

The original research framed this as an affordable alternative to enterprise construction software — photo documentation, daily logs and change orders for small contractors priced out of Procore. That framing is American, it is crowded, and it loses.

The South African framing is different and better:

> ***A South African subcontractor loses money not because they cannot prove what happened on site, but because they missed a contractual notice deadline and had no contemporaneous record to attach when they finally submitted the claim.***

That single sentence is the product. Everything else — the photos, the site diary, the compliance register — exists to serve it.

This matters because it changes what you are selling. "Photo documentation app" is a productivity purchase, competes with CompanyCam, and gets compared on price. "The system that stops you forfeiting claims" is a risk purchase, competes with nothing local, and gets compared to the cost of one lost claim.

The three things the product does

Function	What it prevents	Who feels the pain
Claim notice tracking and generation	Forfeiture of an extension of time or variation claim through late notice	Contractor / subcontractor principal
Contemporaneous site record (diary + photos)	A claim being notified in time but rejected for lack of substantiation	Site agent, foreman, the QS who has to build the claim
Statutory compliance register	Inability to produce documents during a DEL inspection; expired Letter of Good Standing blocking site access or payment	Contractor principal, safety officer, principal contractor

These are not three products. They share one timeline, one project, one evidence store. That integration is the defensibility.

1.2 The architectural inversion

The original design document chained the system as:

```
Photo (evidence) -> Daily Log (narrative) -> Change Order (contract change)
                                          -> Compliance Entry (safety)
```

For this market, invert it:

```
                        CLAIM / NOTICE (the commercial object, with a running clock)
                                     ^
                                     |  substantiated by
                                     |
        SITE DIARY (dated, contractual)  <--  PHOTOS (located, timestamped, integrity-sealed)
                    |
                    +--> COMPLIANCE REGISTER (statutory, separately queryable)
```

The claim is the root object. It has state, deadlines, and a document trail. The diary and photos are its evidence base and also stand alone. The compliance register is a parallel spine that shares the project and date axes but answers to a different audience (an inspector, not a principal agent).

Practical consequence for your schema: a claim is not a child of a diary entry. A claim references diary entries and photos, many-to-many, and can be created without either. Model it as a first-class aggregate root.

1.3 Who actually buys this

Three candidate buyers. They are not equally good and you should pick one before you write code.

Buyer A — the small subcontractor principal

A plumbing, electrical, or general building sub, CIDB Grade 2 to 5, six to forty people, working under a principal contractor on JBCC or under a public client on GCC/NEC.

- **Pain:** loses claims, chases payment, cannot produce compliance documents when the PC demands them, gets kicked off site when the Letter of Good Standing expires.
- **Willingness to pay:** low in absolute terms, but a single recovered claim dwarfs the subscription. Framing matters enormously.
- **Reachability:** hard. Diffuse, unbranded, not on LinkedIn, does not read software reviews.
- **Verdict:** the right end user. The wrong first customer.

Buyer B — the principal contractor / main contractor

A main contractor running a site with eight to twenty subs under them.

- **Pain:** they are legally exposed for the subs' safety compliance under the Construction Regulations and Section 37(2) mandatary agreements. They currently chase this with WhatsApp and email. They also have to assess incoming EOT claims from subs within their own 20-working-day window.
- **Willingness to pay:** materially higher, and they can mandate the tool onto their subs.
- **Reachability:** far easier. Fewer of them, they have identifiable staff, they attend industry events.
- **Verdict:** this inverts your pricing model in a good way. The PC pays; the subs get accounts. This is the "principal contractor pays, subcontractor uses" model and it solves your distribution problem.

Buyer C — the construction health and safety officer / consultant

A SACPCMP-registered CHSO servicing ten to twenty small contractors, currently drowning in Word templates and Dropbox folders.

- **Pain:** the paperwork is their entire job and it is manual.
- **Willingness to pay:** moderate, but they are a **channel**, not just a customer. One CHSO adopting your tool brings fifteen contractors.
- **Verdict:** the best first conversation you can have, whether or not they are the eventual buyer.

Recommendation: talk to Buyer C first to learn the domain, sell to Buyer B, serve Buyer A.

1.4 Competitive landscape

International tools that will show up in your deals

CompanyCam — the dominant photo documentation product globally. In 2026 it restructured pricing into Core at $63/month for one user, Crew at $129 and Scale at $199, eliminating the long-criticised three-user minimum, and expanded beyond photos into estimating, payments, invoicing and document signing after acquiring Beam Finance. Previously it was $79/month minimum for three users. Sources: Manifold's 2026 pricing breakdown, Manifold on the 2026 restructure, G2 pricing, Capterra.

Raken — the dominant daily-reporting product. Covers daily reports, photo documentation, safety observations, toolbox talks, incident reporting and managed checklists. Pricing is quote-based; third-party sources put the ladder around $15 / $37 / $46 per user per month on annual billing, with reviewers flagging year-two increases. Notably its Android app sits around 4.3 stars across roughly 2,400 reviews, close to a full star below iOS, with crash complaints. Sources: Contractor ToolStack review, Software Advice profile, ForConstructionPros, ConstructionBids.ai review.

Others you will meet: Procore, Fieldwire, Buildertrend, Contractor Foreman, JobTread, SafetyCulture, Manifold, FieldFuze, JobNimbus.

Why none of them beat you locally

None of them implement:

- JBCC / NEC / GCC notice regimes and their working-day clocks
- The South African public holiday calendar and the December builders' holiday shutdown
- Construction Regulations 2014 appointment letters and Regulation 4 notification
- COIDA Letter of Good Standing expiry tracking

- CIDB grading and Register of Projects context
- POPIA-aware data handling and local (af-south-1) evidence residency

And they are priced in USD with support in US business hours.

The local landscape

South African safety software skews enterprise and consulting-led — international EHS platforms configured for local legislation by local implementation partners for corporate clients. One such partner makes the gap explicit: the OHS Act, Construction Regulations and COID Act create specific documentation, audit trail and contractor management obligations that international platforms built for other markets do not always address (OVS Solutions guide).

That is a top-heavy market. The Grade 2 sub with a bakkie and six people is served by nobody.

The competitive risk you should actually worry about

Not CompanyCam. It is a local QS or safety consultant with a developer friend, building the same thing, with better domain knowledge than you and worse engineering. Your advantage is that you can ship a properly offline-capable, properly secured mobile platform and they cannot. Their advantage is that they already know what a Regulation 4 notification looks like. Close that gap fast — that is what Part 2 is for.

1.5 Pricing and billing for South Africa

Do not convert the USD tiers

$29 / $79 / $129 converts to roughly R520 / R1,420 / R2,320. The top tier is a very hard sell to a Grade 2–4 contractor.

A shape that fits the market

Tier	Indicative price	Who	Core inclusion
Site	R249–R399 / month	Solo / one active site	Diary, photos, PDF reports, offline
Contractor	R699–R999 / month	Multi-site sub	+ claims register, notice engine, compliance register
Principal Contractor	R2,500–R6,000 / month	Main contractor, per active site	+ subcontractor accounts, rollup dashboard, s37(2) register

The Principal Contractor tier is where the revenue is, and it is priced per active site rather than per user — which is the correct unit for construction, because sites have defined lifespans and crews rotate.

Free subcontractor seats under a paying PC is the growth loop. The sub experiences the product on someone else's account, then buys their own when they move to a site where the PC does not provide it.

Billing mechanics

Stripe does not support South African-registered merchants for Checkout or Connect payouts in ZAR; settlement is restricted to US and EU entities (Stripe global availability, analysis for SA founders, Skydo guide).

You already run **Paystack** in production with payouts approved. That is the answer. Two specific things to build for:

- **Debit order is the dominant SMB recurring instrument in South Africa.** Card-on-file churn from expiry and insufficient funds is brutal at low ticket sizes. Support both.
- **Annual prepay at a discount** works well with contractors who budget per project, and it front-loads cash.

If you later want to sell into other African markets, Paystack already covers Nigeria, Ghana, Kenya and Côte d'Ivoire. If you ever want US customers, use a merchant-of-record such as Paddle or Lemon Squeezy rather than incorporating abroad — roughly 5% + $0.50 per transaction, with tax handled.

Revenue shape

At 300 paying accounts blended at R800/month you are at roughly R2.9m ARR. At 60 principal contractors at R4,000/month plus 400 subs at R700 you are at roughly R6.2m ARR. The PC-led model reaches a viable solo-founder business at roughly one-fifth the customer count. That is the whole argument for Buyer B.

1.6 What to validate before writing code

Ninety days, in this order.

Weeks 1–3: learn the domain. Work through Part 2 of this document with the primary sources open. Read the JBCC advisory notes on extension of time end to end. If you can afford it, buy a copy of the JBCC Principal Building Agreement and read clause 23 and the contract instruction clauses in full.

Weeks 3–6: talk to three CHSOs and five subcontractors. Questions that matter:

- "Walk me through the last time you claimed an extension of time. What did you send, when, and what happened?"
- "Show me the last site diary you kept." (Ask to actually see it. The format tells you the schema.)
- "When last did the Department inspect a site you were on? What did they ask for?"
- "Who chases you for your Letter of Good Standing, and how?"
- "Who pays for the safety file — you or the principal contractor?"

Weeks 6–8: talk to three principal contractors. The single question that decides your business model: *does the PC pay for the sub's compliance tooling, or force the sub to pay?* If the PC pays, build for Buyer B immediately.

Weeks 8–12: build the narrowest possible thing. Not the four-layer platform. One project, JBCC only, site diary plus photos plus the EOT notice clock. Give it to two contractors free. See whether the clock ever fires.

The signal you are looking for: a contractor tells you the notice reminder saved a claim. That is the entire product thesis in one anecdote. If it never happens in three months, the thesis is wrong and you should know that before you have built a compliance module.

PART 2 — THE CONSTRUCTION DOMAIN

This is the part to read twice. Everything in Parts 3 and 4 depends on it.

2.1 How a South African construction project is structured

Software people usually model construction as "client hires builder." It is not that. On any project large enough to matter, there are five to seven distinct parties with distinct contractual roles, and getting them wrong in your data model will make the product feel foreign to users.

The parties

Employer (JBCC) / **Client** (Construction Regulations) / **Employer** (GCC) — the party paying for the works. Note the terminology collision immediately: the Construction Regulations call this party the *client* and impose specific health-and-safety duties on them, while JBCC calls the same party the *employer*. Your UI should use the term appropriate to the context.

Principal Agent — under JBCC, the single party appointed by the employer with authority to administer the contract: issue contract instructions, certify payment, grant or refuse extensions of time, and certify practical completion. Usually the architect on a building project. **This is the single most important external party in your product**, because almost every notice your users send is addressed to the principal agent, and every response they wait for comes from the principal agent.

Agent (Construction Regulations sense) — a completely different thing. Where a construction work permit is required, the client must appoint a competent person in writing as an agent to manage health and safety on the project, and that person must be registered with a statutory body approved by the Chief Inspector. The SACPCMP is the approved body. Do not conflate the JBCC principal agent with the Construction Regulations agent — they are different appointments under different instruments and are often different people.

Quantity Surveyor (QS) — values the works, prices variations, assesses claims. On claims, the QS is often the person who actually builds the substantiation. A QS is frequently the person who will evaluate your product on behalf of a contractor.

Principal Contractor — the main contractor with overall responsibility for the site, including health and safety coordination of subcontractors under the Construction Regulations.

Contractor / Subcontractor — your primary user base. Note that JBCC distinguishes *nominated*, *selected* and *domestic* subcontractors with different contractual chains. A domestic sub contracts with the main contractor; a nominated sub is imposed by the principal agent. This affects who they send notices to.

Construction Health and Safety Officer (CHSO) / Agent (CHSA) / Manager (CHSM) — SACPCMP-registered safety roles. The CHSO is your channel play.

The consequence for your data model

A project needs a **party register** — a table of the parties on that project with their role, contact details and (critically) which notices go to which of them. A notice sent to the wrong party may be no notice at all.

```
project_parties
  id, project_id, party_type, org_name, contact_name,
  email, phone, is_notice_recipient, appointed_at, appointment_document_id
```

`party_type` is an enum that must be contract-form aware: `employer`, `principal_agent`, `quantity_surveyor`, `principal_contractor`, `contractor`, `subcontractor`, `cr_client`, `cr_agent`, `chso`, `chsa`, `chsm`.

2.2 The standard-form contracts

Four families dominate South African construction. Your product must know which one a project is under, because the notice regimes differ substantially.

JBCC — Joint Building Contracts Committee

Dominant in **private sector building work**. The core document is the **Principal Building Agreement (PBA)**. There are matching subcontract agreements (Nominated/Selected Subcontract Agreement, Domestic Subcontract Agreement),

a Minor Works Agreement for small projects, and a Contract Data document where project-specific values live (dates, penalties, the builders' holiday period).

This is your **starting contract form**. Most small subcontractors in private building work are on JBCC or on a bespoke contract that borrows JBCC's shape.

Official body: jbcc.co.za. Their published advisory notes are free and are the single best learning resource available — see the EOT advisory note and advisory note on clause 23.

GCC — General Conditions of Contract for Construction Works

Published by SAICE (South African Institution of Civil Engineering). Dominant in **public sector civil and infrastructure work** — municipalities, water boards, roads.

Notably, GCC does not define normal versus abnormal weather; the crossover point is meant to be agreed between the parties, informed by the contractor's obligation to have inspected the site and be knowledgeable about local climatic conditions (analysis of GCC weather provisions).

NEC — NEC3 / NEC4 Engineering and Construction Contract

Used extensively by large South African public entities and state-owned enterprises. NEC's model is fundamentally different: **early warnings** and **compensation events**, with a notification time bar that is famously unforgiving. If you support NEC, the early warning register is itself a product feature — it is a live, shared document that both parties are contractually obliged to maintain.

FIDIC

International standard forms, used on large infrastructure and donor-funded projects. Claim notice provisions are strict and time-barred.

Design implication

Contract form is a first-class property of a project, set at creation, and it drives:

- which notice types exist
- the deadline arithmetic for each
- the template used to generate the notice document
- the vocabulary shown in the UI

```
contract_forms
  id, code, name, edition, effective_from
  -- e.g. ('jbcc_pba_6_2', 'JBCC Principal Building Agreement', '6.2', '2018-05-01')

notice_rules
  id, contract_form_id, notice_type, clause_reference,
  trigger_description, period_value, period_unit, period_basis,
  consequence_of_failure, template_id
  -- period_unit: 'working_days' | 'calendar_days'
  -- period_basis: 'from_awareness' | 'from_cessation' | 'from_receipt'
```

Versioning `contract_forms` by edition is not optional. Clause numbering changes between editions and a generated notice citing the wrong clause is worse than no citation.

2.3 Contract instructions and variations

Under JBCC there is no single "change order." There are three distinct concepts and your model must separate them:

Contract Instruction (CI) — an instruction issued by the principal agent to the contractor. It is the *event*. It may or may not have cost or time consequences.

Variation / adjustment of the contract value — the *money* consequence. Priced by the QS, usually against contract rates where they apply, otherwise on a fair valuation or cost-plus basis.

Revision of the date of practical completion (EOT) — the *time* consequence. Handled under an entirely separate clause with its own notice regime.

A single CI for additional brickwork may generate a variation with no EOT, an EOT with no variation, both, or neither.

The verbal instruction problem

This is a large, under-served pain point specific to how sites actually work. A site agent is told verbally to do something extra. They do it. Nobody records it. At final account the QS finds no instruction and the work is not paid for.

The mitigation is a **confirmation of verbal instruction (CVI)** — a written note back to the principal agent recording what was instructed, when, and by whom, giving them the opportunity to dispute it. Most contracts allow this and some require it within a set period.

This is a small feature with disproportionate value. One screen: who instructed, what, when, which project. Generates a formal CVI letter, emails it to the principal agent with delivery evidence, files it against the project. Build this early — it demonstrates value in week one of a trial.

Schema sketch

```
contract_instructions
  id, project_id, ci_number, issued_by_party_id, issued_at,
  received_at, description, source, -- 'written' | 'verbal_confirmed'
  cvi_sent_at, cvi_document_id, disputed, created_by, created_at

variations
  id, project_id, contract_instruction_id, variation_number,
  description, valuation_basis, labour_amount, material_amount,
  plant_amount, subcontract_amount, preliminaries_amount,
  markup_pct, submitted_at, status, certified_amount, certified_at
```

Keep the cost breakdown itemised. When a variation is disputed, the argument is almost always about one component, and an unitemised lump sum is indefensible.

2.4 Extension of time: the notice regime

This is the core of the product. Read this section carefully.

Under the JBCC Principal Building Agreement, clause 23 governs revision of the date for practical completion. The mechanism has three stages, each with its own clock.

Stage 1 — entitlement

Clause 23.1 lists circumstances entitling the contractor to a revision **without** an adjustment of the contract value. Clause 23.2 lists circumstances entitling a revision **with** cost. Adverse weather sits in clause 23.1.1 — an extension of time but no money.

Stage 2 — notice (the fatal deadline)

The contractor must give the principal agent reasonable and timeous notice of the circumstances giving rise to the delay, and **within 20 working days** from when the contractor became aware, or ought reasonably to have become aware, of the potential delay, notify the principal agent of its intention to submit a claim for revision of the date of practical completion.

If this notice is not given within that period, the contractor forfeits the claim entirely — regardless of how real the delay was. Where the contractor fails to adhere to these provisions, the claim is unenforceable. Sources: Cox Yeats on clause 29/23 EOT claims, Hahn & Hahn: "You mentioned it… but did you notify?".

Stage 3 — quantification

Once the delay has ceased, the contractor has **40 working days** from when it is able to quantify the delay to submit the claim details to the principal agent, including the extension period claimed in working days and the calculation thereof. Sources: JBCC advisory note on EOT, Le Roux Consulting guide.

Stage 4 — the principal agent's clock

The principal agent has **20 working days** from receipt of the claim to grant it in full, in part, or reject it. **If the principal agent fails to respond, the claim is deemed rejected**, at which point the contractor may issue a notice of disagreement and the ruling can be reviewed. Sources: Agora Africa on the principal agent's role, JBCC advisory note.

Subcontracts run tighter

Subcontract forms compress these periods. The older JBCC Series 2000 Edition 5.0 subcontract required the subcontractor to notify the contractor of its intention to claim **within 15 working days** (Lexology analysis). Verify the current subcontract edition before implementing — but the principle holds: **subcontractor deadlines are shorter than main contractor deadlines**, and your primary user is the subcontractor.

The three-step framing practitioners use

Local claims practitioners describe successful EOT claims as three steps: establish entitlement through timeous notice and adherence to the notice provisions; quantify by demonstrating the impact of the delay on the critical path of the current programme; and provide corroborating evidence and contemporaneous documents to support key facts and assumptions (Le Roux Consulting).

Map that directly onto your product:

Practitioner step	Your feature
Entitlement through timeous notice	The notice engine
Quantify against the programme	Programme import and delay-day calculation
Contemporaneous documents	The site diary and photo evidence store

Step 2 is the hardest and involves critical-path analysis. **Do not build it in v1.** Let the user enter the claimed days manually and attach their own programme extract. Build steps 1 and 3 properly and you have a product.

The state machine

```
POTENTIAL_DELAY_IDENTIFIED
   -> (clock A: 20 working days from awareness)
NOTICE_OF_INTENTION_SENT
   -> delay ceases
   -> (clock B: 40 working days from ability to quantify)
CLAIM_SUBMITTED
   -> (clock C: 20 working days for principal agent)
GRANTED_FULL | GRANTED_PARTIAL | REJECTED | DEEMED_REJECTED
   -> NOTICE_OF_DISAGREEMENT_SENT
   -> DISPUTE

Terminal failure states:
NOTICE_LAPSED       (clock A expired without notice)
CLAIM_LAPSED        (clock B expired without submission)
```

`NOTICE_LAPSED` should be a loud, unmissable state in the UI. It is the thing the product exists to prevent, and when it happens anyway the user needs to know immediately, not at final account.

2.5 Weather as a claim ground

Weather deserves its own section because it is the most common EOT ground and because it drives the design of your site diary.

The JBCC position is contractor-friendly — but only with records

Adverse weather is a listed clause 23.1.1 event. The JBCC has clarified that there is **no onus to prove the weather was "exceptional"**; the contextual usage of "adverse weather" refers to any circumstance related to weather that inhibits progress towards practical completion (academic analysis of JBCC claim events, SciELO).

That is a low entitlement bar. But the same JBCC advisory note is explicit that **the mere fact that it rained on a particular occasion is not grounds for an extension** — what matters is the effect. The advisory gives useful examples: light overnight rain can qualify if it affects foundation excavation or makes high-rise structural steel slippery and dangerous; heavy rain during working hours at completion stage may not qualify if it causes no delay to practical completion; soggy ground conditions from earlier rainfall may only show their consequential delay a day or two later and can persist (advisory note 23.0).

The design consequence

A weather icon is worthless as evidence. Your daily weather capture must record *effect*, not condition:

Field	Why
Condition (rain / wind / heat / cold)	The trigger
Which activities were stopped or slowed	Establishes the link
Hours lost per activity	Quantum
Was the affected activity on the critical path?	Determines whether the delay is claimable at all
Delayed-effect flag	Captures the "ground was still wet on Tuesday from Sunday's rain" case, which is where most claims are lost

The delayed-effect flag matters more than it looks. The advisory explicitly recognises consequential delay evident only a day or two later. A diary that only records "rain: yes" on the day it rained cannot substantiate the two subsequent lost days.

GCC is different

Under GCC the crossover from normal to abnormal weather is not defined in the contract and should be agreed between the parties, established through the contractor's obligation to inspect the site and be knowledgeable about local climatic conditions. Rainfall that could not have been foreseen from available rainfall records for the area, by intensity or frequency, would be abnormal and entitle an extension — and the actual delays caused must be recorded and agreed by the parties (Lexology on GCC). Some employers avoid the argument by specifying a formula or a stated number of rain days at the contractor's risk.

Product feature: allow a project to store a rainfall baseline (either a contractual figure or regional historical data) and flag days that exceed it. Rainfall data can come from the South African Weather Service, but a contractor may also be contractually required to keep site rainfall records themselves — the site diary should support a manual rainfall reading in millimetres.

Practitioner guidance suggests the contractor should provide equipment as part of site establishment to record abnormally high or low temperatures, cloudbursts and the quantum of rain over 24 hours, and extreme wind speed or gusts, noting the time of day (Plumbing Africa on weather and subcontractor obligations).

A note on regional patterns: Gauteng, KwaZulu-Natal and Mpumalanga receive most rain in summer; the Western Cape is wetter in winter (regional overview). Your baseline logic must be region-aware, not national.

2.6 Loss and expense, standing time and plant

Two fields missing from most daily-log products that materially affect claim value:

Plant on site, split working versus standing. Standing time on hired plant is a recoverable cost in a clause 23.2-type claim. A crane idle for three days because of a late instruction is real money that nobody records. Your diary should list plant items on site with hours working and hours standing.

Labour by trade and by subcontractor, not a headcount. On a multi-sub site, whose people were present matters for claim substantiation, for productivity loss arguments, and for Section 37(2) safety obligations.

```
diary_labour
  id, diary_entry_id, trade, subcontractor_party_id,
  headcount, hours_worked, hours_standing, notes

diary_plant
  id, diary_entry_id, plant_description, plant_ref,
  hours_working, hours_standing, is_hired, hire_rate_basis
```

2.7 The site diary as a contractual document

Call it a **site diary**, not a "daily log." That is what the market calls it and what contracts refer to.

The site diary is not an internal note. On most projects it is a recognised project document, sometimes contractually required, and it is the first thing a QS or attorney asks for when a claim is being built.

Minimum fields for a claim-grade diary

- Date, project, and author (with role)
- Weather: condition, effect, hours lost, affected activities, delayed-effect flag, rainfall reading
- Labour by trade and subcontractor with hours
- Plant on site, working versus standing
- Work completed, by area or work section
- Materials delivered and used, with supplier and quantity
- Deliveries expected but not received (a delay ground in its own right)
- Instructions received that day (written or verbal)
- Drawings and revisions received
- Visitors to site (relevant to both claims and safety)
- Delays and disruptions, with a "potentially claimable" flag
- Attached photos
- Signature or acknowledgement

The two-minute problem

Your original design goal was a diary completable in under two minutes. That goal fights directly against claim-grade records. Resolve it with **two-tier entry**:

Tier 1 (default, fast): date, crew count, weather condition, work completed, photos. Sixty seconds.

Tier 2 (escalated): triggered when the user flags weather as having stopped work, records a delay, or marks something as potentially claimable. The app then asks the three or four questions that make the entry usable in a claim, and offers to start the notice clock.

The escalation is the product. Most days are Tier 1. The days that matter are Tier 2, and on those days the foreman will spend three extra minutes if you tell them why.

2.8 Health and safety: OHSA and the Construction Regulations 2014

The statutory stack

Occupational Health and Safety Act 85 of 1993 (OHSA) — the parent statute. Section 37(2) is the provision governing mandataries; a written agreement between an employer and a mandatary (contractor) arranging their respective duties is what shifts liability. **Section 37(2) mandatary agreements have no United States analogue** and are a genuine documentation pain point.

Construction Regulations 2014 — made under section 43 of OHSA, applying to all persons involved in construction work (PolicyVault summary, full text PDF, ASAQS guidelines PDF).

Regulation 3 — construction work permit

A client intending to have construction work carried out must, at least 30 days before work commences, apply to the provincial director in writing for a construction work permit where the works contract exceeds the threshold. The regulation as published set this at a works contract value equal to or exceeding **thirteen million rand or CIDB grading level 6** (Polity summary). The Chief Inspector subsequently issued a notice granting temporary exemption and raising the trigger to a works contract value **exceeding forty million rand** for construction work starting on or after 7 February 2017 (Labour Guide notice, SAIOSH notice).

Where a permit is required, the client **must** appoint a competent person in writing as an agent, and that agent must be registered with the statutory body approved by the Chief Inspector — the SACPCMP (Construction Regulations text, Polity).

Verify the current threshold before building anything that depends on it. It has been changed by notice at least once.

Regulation 4 — notification of construction work

This is where your users live. A contractor who intends to carry out any construction work other than work contemplated in regulation 3(1) must, **at least 7 days before that work is to be carried out**, notify the provincial director (Regulation 4 text).

Small contractors and subcontractors sit almost entirely under Regulation 4, not Regulation 3. It is a repeated, per-

project obligation with a legal deadline. That is a product hook: **project created → notification due in 7 days → generate the form → track submission and acknowledgement.**

The other documentation obligations

The Construction Regulations run heavily on written records:

- **Written appointments** — construction manager, construction supervisor, and a series of competent-person appointments for specific activities (excavation, scaffolding, fall protection, temporary works, and so on). These expire, get superseded, and get requested during inspections.
- **Health and safety specification** (client duty) and **health and safety plan** (contractor duty).
- **Baseline and activity-specific risk assessments**, reviewed periodically.
- **Fall protection plan** where working at height.
- **Section 37(2) mandatary agreements** between principal contractor and each contractor.
- **Induction records, toolbox talks, PPE issue registers.**
- **Incident records.**

Scale context: the Department has cited an average of roughly 12,500 construction sites in a financial year, with the industry employing around 1.46 million workers (DEL presentation PDF).

Practitioner commentary notes the regulations are generally complied with by larger contractors but **less so by SMME contractors, who often have limited cash resources** (Plumbing Africa). That is your market and your opportunity, but also a warning: they are non-compliant partly because compliance costs money they do not have. Price accordingly.

Schema sketch

```
appointments
  id, project_id, appointment_type, appointee_name, appointee_id_ref,
  appointed_by_party_id, appointed_at, expires_at, superseded_by,
  document_id, acknowledgement_signed_at

compliance_records
  id, project_id, record_type, -- toolbox_talk | ppe_check | hazard |
                               -- induction | incident | inspection
  occurred_at, topic, description, mitigation, attendees_json,
  photo_ids, created_by, created_at

statutory_documents
  id, org_id, project_id, doc_type, -- letter_of_good_standing |
                                    -- cidb_certificate | hs_plan |
                                    -- risk_assessment | s37_2_agreement
  issued_at, expires_at, reference, file_id, status
```

The `expires_at` column is where a lot of the product's day-to-day value sits. Expiry alerting on statutory documents is unglamorous and people will pay for it.

2.9 COIDA and the Letter of Good Standing

Compensation for Occupational Injuries and Diseases Act 130 of 1993 (COIDA) is South Africa's workplace injury compensation system, funded entirely by employer contributions.

Key mechanics:

- Registration is mandatory for any employer with one or more employees.
- Registration is done through the Compensation Fund's online portal at cfonline.labour.gov.za, producing a **CF reference number** (registration guide, ClearComply 2026 guide).
- The **Return of Earnings (ROE)** is an annual declaration of total employee earnings for the assessment period, which runs 1 March to end February. The online ROE deadline is **30 June**, filed via roe.labour.gov.za; paper submissions carry an earlier 31 March deadline. The Notice of Assessment must be paid within 30 days of receipt (ClearComply).
- Once the assessment is paid, the employer can obtain a **Letter of Good Standing**, valid for 12 months, which most B2B clients, government tenders and the Central Supplier Database require before they will contract or pay (registration guide).
- Late ROE submission triggers a penalty plus interest, and the Letter of Good Standing is suspended until paid.
- Without registration, an employer loses the civil liability protection COIDA provides — an injured employee can sue directly, and the Fund can recover compensation it pays from the employer (ClearComply).

Why this matters commercially

The Letter of Good Standing is a gating document. A subcontractor without a current one can be barred from site and blocked from payment. It expires annually. It depends on an annual filing with a fixed deadline. Contractors lose work over this.

An expiry tracker with escalating reminders at 60, 30 and 7 days, plus an ROE deadline reminder each April, is a small feature with a very direct link to money. It is also an excellent free-tier hook: give it away, capture the account, upsell the claims module.

Note: sources differ on the registration window (7 versus 14 days from commencing employment) and on Letter of Good Standing renewal timing. Verify with the Compensation Fund directly before encoding any of this as an in-product deadline.

2.10 CIDB grading and the Register of Projects

The **Construction Industry Development Board (CIDB)** is a statutory body established under the CIDB Act 38 of 2000. It maintains the **Register of Contractors (RoC)**, which grades contractors 1 to 9 according to financial and works capability, and a separate **Register of Projects** (CIDB overview, Construction Registers Service).

Key facts:

- The Construction Industry Development Regulations of 2004, as amended, **bind public sector clients to award construction contracts only to CIDB-registered contractors** (CIDB).
- A grade is one digit and two letters — for example **4GB** — where the digit is the grading level determining the maximum project value (tender value limit) and the letters are the class of works, such as General Building or Civil Engineering (CivilSure explainer).
- **Grade 1 has no qualifying criteria.** Grades 2 to 9 require the contractor to meet **financial and works capability (track record) criteria** (CIDB, CIDB comprehensive guide PDF).
- Contractors apply for upgrades as they complete larger projects and strengthen financially.
- Subcontractors do not currently need to register; joint ventures do not register but all JV partners must be individually registered (CIDB).

The commercial angle worth investigating

Works capability is assessed on completed project track record. A tool that maintains a clean, dated, evidenced record of every project a contractor completes — value, class of works, dates, client reference, photographic evidence of the completed works — is producing exactly the substantiation an upgrade application needs.

That reframes the value proposition from "insurance against disputes" to **"this gets you to the next grade, which lets you tender for bigger work."** That is a growth purchase, not a risk purchase, and growth purchases are easier to sell.

Verify before you build on this. Read the current CIDB Grade 2 to 9 application form and instructions and the comprehensive guide to see exactly what evidence the CIDB accepts. If they require client-certified completion certificates rather than contractor-held records, the feature is weaker — still useful for assembling the pack, but not the whole story. A caution the CIDB itself publishes: contractors remain accountable for all information submitted in their company name under Regulation 30 of the CID Regulations (CIDB). Do not build anything that could be read as generating a track record rather than recording one.

2.11 SACPCMP and the CHSO channel

The **South African Council for the Project and Construction Management Professions (SACPCMP)** is the statutory body approved by the Chief Inspector for registration of construction health and safety practitioners (Polity on the approval).

Three relevant registration categories: Construction Health and Safety Agent (CHSA), Construction Health and Safety Manager (CHSM), and Construction Health and Safety Officer (CHSO).

For your purposes, the CHSO community is:

- A **distribution channel** — one CHSO serves many small contractors.
- A **credibility source** — a product endorsed by registered professionals is taken seriously.
- A **domain teacher** — three conversations with an experienced CHSO will teach you more about the compliance

layer than a month of reading.

Also relevant: **SAIOSH** (South African Institute of Occupational Safety and Health) publishes notices and industry updates and is a good place to track regulatory change (saiosh.co.za).

2.12 Data protection and electronic evidence

Two statutes that directly constrain your design.

POPIA — Protection of Personal Information Act 4 of 2013

Site photos contain workers' faces. That is personal information, and you will be processing it on behalf of contractors who have no capacity to be a compliant responsible party.

The critical constraint: **section 26 of POPIA defines "special personal information"** to include religious or philosophical beliefs, race or ethnic origin, trade union membership, political persuasion, health or sex life, **or biometric information**, and criminal behaviour. Processing special personal information is **prohibited** unless the exceptions in sections 27 to 33 apply or the Regulator has granted an authorisation under section 27(2) (POPIA section 26 text, POPIA overview, Usercentrics overview).

Penalties run to fines up to R10 million and imprisonment up to 10 years (POPIA overview, biometric compliance analysis).

Direct consequence for your roadmap: the original stack document proposes face recognition under the AI services layer and biometric attendance options in Phase 2, qualified only as requiring "appropriate consent and controls." That understates the position materially. Biometric information is a prohibited category by default with a higher bar than ordinary consent.

Recommendation: cut biometrics from the roadmap entirely. Use QR or NFC worker attendance. It is the same feature at a fraction of the legal exposure, and it does not require you to become an expert in section 27 authorisations on behalf of customers who will not read your consent copy.

Other POPIA obligations to design for from day one: the eight conditions for lawful processing in Chapter 3, security safeguards under section 19, purpose limitation and further-processing limitation under section 15, and the cross-border transfer rules. Storing evidence in **af-south-1** rather than us-east-1 simplifies the last one and is a genuine selling point against international competitors.

Regulator: the Information Regulator of South Africa.

ECTA — Electronic Communications and Transactions Act 25 of 2002

Your entire "evidentiary weight" claim rests on this statute, not on device timestamps. ECTA governs the admissibility and evidential weight of data messages, and sets out the factors a court considers — including the reliability of the manner in which the data message was generated, stored and communicated, the reliability of the manner in which the integrity of the information was maintained, and the manner in which its originator was identified.

Read section 15 at source before writing any marketing copy about evidence. But the shape of the test tells you exactly what to build, which is covered in section 3.5 below.

The honest position on device metadata

A device timestamp and GPS coordinate are trivially spoofable — mock location applications, rooted or jailbroken devices, changed system clocks. The original design document's claim that timestamps are "not manually editable after the fact" and therefore give photos evidentiary weight is **not true as specified** and should not appear in your marketing.

What gives evidence weight is the reliability of the system that received and stored it. That is a server-side property, not a device property.

2.13 Glossary

Use these terms in the product. Using American equivalents will cost you credibility instantly.

Use this	Not this
Site diary	Daily log
Contract instruction (CI)	Change order
Variation	Change order (cost)
Extension of time (EOT) / revision of the date for practical completion	Schedule extension
Principal agent	Project manager / architect
Employer (JBCC) / Client (Construction Regulations)	Owner
Principal contractor	General contractor (GC)
Site agent	Superintendent
Practical completion	Substantial completion
Snag list / practical completion list	Punch list
Preliminaries and generals (P&Gs)	General conditions
Bill of quantities (BOQ)	Schedule of values
Letter of good standing	(no equivalent)
Toolbox talk	(same)
Section 37(2) mandatary agreement	(no equivalent)
Notification of construction work	(no equivalent)
CIDB grading	(no equivalent)
Bakkie	Pickup truck

PART 3 — SYSTEM DESIGN

3.1 Domain model

The core aggregate roots, in dependency order.

```
organisation (tenant)
  └── users, roles
  └── statutory_documents (LGS, CIDB cert, insurances)
  └── projects
        ├── contract (form, edition, contract data values)
        ├── project_parties (employer, PA, QS, PC, subs, CHSO)
        ├── diary_entries
        │     ├── diary_labour
        │     ├── diary_plant
        │     ├── diary_materials
        │     ├── diary_weather
        │     └── evidence_links -> photos
        ├── photos (evidence)
        ├── contract_instructions
        │     └── variations
        ├── claims  ◄── the commercial spine
        │     ├── claim_events (state transitions)
        │     ├── claim_deadlines (the clocks)
        │     ├── evidence_links -> diary_entries, photos, documents
        │     └── generated_documents (notices, submissions)
        ├── compliance_records
        ├── appointments
        └── generated_reports
```

The claims table

```
CREATE TABLE claims (
    id                  uuid PRIMARY KEY,
    org_id              uuid NOT NULL REFERENCES organisations(id),
    project_id          uuid NOT NULL REFERENCES projects(id),
    claim_ref           text NOT NULL,          -- user-visible, per project
    claim_type          claim_type_enum NOT NULL,
        -- 'eot' | 'variation' | 'loss_and_expense' | 'compensation_event'
    contract_form_id    uuid NOT NULL REFERENCES contract_forms(id),

    -- entitlement
    clause_relied_on    text,                   -- e.g. '23.1.1'
    trigger_description text NOT NULL,
    awareness_date      date NOT NULL,          -- starts clock A
    delay_ceased_date   date,                   -- starts clock B
    quantifiable_date   date,                   -- alternative basis for clock B

    -- quantum
    days_claimed        integer,
    cost_claimed_cents  bigint,
    currency            char(3) NOT NULL DEFAULT 'ZAR',

    status              claim_status_enum NOT NULL,
    is_critical_path    boolean,

    created_by          uuid NOT NULL,
    created_at          timestamptz NOT NULL DEFAULT now(),
    updated_at          timestamptz NOT NULL DEFAULT now(),

    UNIQUE (project_id, claim_ref)
);

CREATE TABLE claim_deadlines (
    id              uuid PRIMARY KEY,
    claim_id        uuid NOT NULL REFERENCES claims(id) ON DELETE CASCADE,
    deadline_type   text NOT NULL,   -- 'notice_of_intention'
                                     -- 'claim_submission'
                                     -- 'pa_response'
    rule_id         uuid REFERENCES notice_rules(id),
    basis_date      date NOT NULL,
    due_date        date NOT NULL,           -- computed, stored
    computed_at     timestamptz NOT NULL,
    calendar_id     uuid NOT NULL REFERENCES working_day_calendars(id),
    satisfied_at    timestamptz,
    satisfied_by_document_id uuid,
    lapsed_at       timestamptz
);

CREATE INDEX idx_claim_deadlines_due
    ON claim_deadlines (due_date)
    WHERE satisfied_at IS NULL AND lapsed_at IS NULL;
```

Store `due_date`, do not compute it on read. Two reasons: the calculation depends on a calendar that can change (a new public holiday is proclaimed, the builders' holiday period is set in contract data), and you need an audit record of what the system told the user and when. Recompute on calendar change and record the recomputation as a `claim_event`.

Evidence linking

Claims reference evidence many-to-many, and evidence can belong to multiple claims:

```
CREATE TABLE evidence_links (
    id              uuid PRIMARY KEY,
    org_id          uuid NOT NULL,
    subject_type    text NOT NULL,   -- 'claim' | 'contract_instruction'
                                     -- | 'compliance_record'
    subject_id      uuid NOT NULL,
    evidence_type   text NOT NULL,   -- 'photo' | 'diary_entry'
                                     -- | 'document' | 'message'
    evidence_id     uuid NOT NULL,
    relevance_note  text,
    linked_by       uuid NOT NULL,
    linked_at       timestamptz NOT NULL DEFAULT now(),
    UNIQUE (subject_type, subject_id, evidence_type, evidence_id)
);
```

Polymorphic joins are unfashionable but correct here — the alternative is six near-identical join tables.

3.2 The notice deadline engine

This is the single most valuable and most dangerous component in the system. Valuable because nothing else does it. Dangerous because if it tells a contractor their deadline is Friday and it is actually Wednesday, you have caused them direct financial loss.

Working-day arithmetic

The JBCC periods are expressed in **working days**, which means:

1. Exclude Saturdays and Sundays.
2. Exclude South African public holidays.
3. Handle the **Public Holidays Act 36 of 1994 Sunday rule** — where a public holiday falls on a Sunday, the following Monday is a public holiday.
4. Handle the **builders' holiday / annual shutdown**. The South African construction industry shuts down for roughly three to four weeks from mid-December. JBCC Contract Data typically specifies this period for the project, and it is excluded from working-day counts. **This is project-specific, not national** — it must be a contract data field, not a hard-coded constant.
5. Handle the contract's own definition of a working day, which may differ.

South African public holidays (fixed-date)

Date	Holiday
1 January	New Year's Day
21 March	Human Rights Day
27 April	Freedom Day
1 May	Workers' Day
16 June	Youth Day
9 August	National Women's Day
24 September	Heritage Day
16 December	Day of Reconciliation
25 December	Christmas Day
26 December	Day of Goodwill

Plus **Good Friday** and **Family Day** (the Monday following Easter Sunday), which move annually and require Easter computation.

Additionally, the President may proclaim ad hoc public holidays (election days have been proclaimed in the past). **Your calendar must be a database table you can update, not a code constant.**

Design

```
CREATE TABLE working_day_calendars (
    id          uuid PRIMARY KEY,
    org_id      uuid,          -- NULL = system default
    project_id  uuid,          -- NULL = org-wide
    name        text NOT NULL,
    country     char(2) NOT NULL DEFAULT 'ZA'
);

CREATE TABLE non_working_days (
    id          uuid PRIMARY KEY,
    calendar_id uuid NOT NULL REFERENCES working_day_calendars(id),
    day         date NOT NULL,
    reason      text NOT NULL,  -- 'public_holiday' | 'builders_holiday'
                                -- | 'site_shutdown'
    source      text,
    UNIQUE (calendar_id, day)
);
```

Resolution order: project calendar → org calendar → system default. Materialise the union into a lookup set per project rather than doing three-way resolution on every calculation.

The calculation

```
// AddWorkingDays returns the date N working days after basis,
// per the supplied calendar. Day 0 is the basis date itself;
// counting starts the following day.
func AddWorkingDays(basis time.Time, n int, cal Calendar) time.Time {
    d := basis
    remaining := n
    for remaining > 0 {
        d = d.AddDate(0, 0, 1)
        if cal.IsWorkingDay(d) {
            remaining--
        }
    }
    return d
}
```

Ambiguity you must resolve with a professional, not guess at: does "within 20 working days from the date on which the contractor became aware" include the awareness date itself? Different practitioners count differently. Get this decided by a QS or construction attorney, document the decision in the code, and — critically — **build in a safety margin in the user-facing alerting.** Alert at 10 days, 15 days and 18 days. Never let the first warning be on the due date.

Timezone

Everything in `Africa/Johannesburg`. You already have `timefmt.go` in `events-db` from fixing the TIMESTAMPTZ/SAST email bug — reuse that approach. Store `timestamptz`, render in SAST, and be explicit in generated documents about which timezone a timestamp is in.

Alerting

```
T-10 working days   in-app + push
T-5  working days   in-app + push + email
T-2  working days   in-app + push + email + SMS (optional paid)
T-0                 escalate to org owner as well as the site user
T+1                 mark LAPSED, notify org owner, log prominently
```

The escalation to the org owner matters. The site agent who misses a notice may not tell the boss. The system should.

The liability question

You are building a system that gives quasi-legal deadline information. Protect yourself and your users:

- Show the clause reference and the source of every computed deadline, so a user can check it.
- Show the calendar used and let the user inspect the excluded days.
- Never present a deadline as authoritative. Present it as *the system's calculation based on the contract form and calendar you selected*, with a prompt to confirm against the contract.
- Log every deadline computation and every notification with a timestamp. If a customer ever claims your product cost them a claim, that log is your defence.
- Get your terms of service reviewed by an attorney before launch.

3.3 Evidence integrity

The ECTA test is about the reliability of generation, storage, communication and integrity maintenance. Design to that test.

What to build

1. Dual timestamps. Record the device timestamp *and* the server receipt timestamp, and show both. When they diverge significantly (offline sync), show the gap honestly rather than hiding it. A photo taken offline on Tuesday and synced Thursday should say exactly that.

2. Content hashing at capture. Compute a SHA-256 of the original image bytes **on the device, before upload**. Store it with the record. Recompute server-side on receipt and compare. This proves the file has not changed between capture and storage.

3. A per-project hash chain. Each evidence record stores the hash of the previous record in the project, forming a

chain:

```
ALTER TABLE photos ADD COLUMN content_sha256   bytea NOT NULL;
ALTER TABLE photos ADD COLUMN chain_prev_hash  bytea;
ALTER TABLE photos ADD COLUMN chain_hash       bytea NOT NULL;
-- chain_hash = SHA256(content_sha256 || server_received_at ||
--                     project_id || chain_prev_hash)
```

This makes retroactive insertion or deletion detectable. It is not a blockchain and does not need to be — it needs to make tampering evident, and a chain anchored by a periodically published root hash does that adequately.

4. Periodic anchoring. Once a day, compute the chain head per project, sign it with a server key, and store it in a separate append-only log (a different table, ideally a different database or an S3 object lock bucket). This is what turns "we hashed things" into "we can demonstrate the record existed on this date."

5. Device attestation. Play Integrity API on Android and DeviceCheck / App Attest on iOS. Attest at capture, store the verdict. This does not make spoofing impossible but it makes it detectable and it is exactly the kind of "reliability of the manner of generation" factor ECTA points at.

6. Mock location detection. Android exposes `isFromMockProvider` on location fixes. Record it. A photo with a mock-provider location should be flagged in the record, not silently accepted.

7. Immutable originals. Store the original bytes and never modify them. Markup, annotation and watermarking produce **derived** files with their own records referencing the original. The original must always be retrievable unaltered.

8. Append-only audit log. Every read, export and share of evidence, logged. Who, when, what, from where.

What to say in marketing

Not "tamper-proof." Say: *every photo is hashed at capture, sealed into a per-project chain, timestamped by our servers independently of the device, and every access is logged* — and then explain that this is designed to address the factors a court considers under ECTA when weighing electronic evidence. That is defensible. "Tamper-proof" is not.

3.4 Offline-first sync

You have solved this before for door scanning in `EventsStudio` with a local SQLite manifest and delta sync. The construction case is harder in one specific way: **it is write-heavy with large binary payloads**, where ticket scanning is read-heavy with small writes.

Architecture

```
Capture  ->  SQLite (local, authoritative until synced)
             + file in app-private storage
         ->  outbox table (queued operations, ordered)
         ->  sync worker
               - metadata first (small, fast, cheap)
               - media second (large, deferrable, resumable)
         ->  server: idempotent upsert by client-generated UUID
         ->  reconcile: mark local rows synced, retain for N days
```

Non-negotiables

Client-generated UUIDs. Every record gets its ID on the device. This makes the sync idempotent — retrying an upload can never create a duplicate. You already do this pattern.

Separate metadata and media queues. A diary entry is a few kilobytes; its photos are tens of megabytes. The diary entry must reach the server even when the photos cannot. The record referencing pending media should be visible server-side with the media in a `pending` state.

Resumable, chunked uploads. S3 multipart with a persisted upload ID, so a dropped connection resumes rather than restarts. On a 3G connection in a valley, restart-from-zero is the difference between a working product and a deleted app.

WiFi-preferred by default. This is a South African product requirement, not a nicety. Mobile data is expensive and your users are on prepaid. Default: queue media, upload on WiFi or on charge. Provide an explicit "upload now on mobile data" button with a size estimate ("Upload 47 photos, approximately 92 MB?"). A foreman who burns R80 of airtime uploading a day's photos will uninstall the app that evening.

On-device compression before queueing, not after upload. Resize to a sensible long edge (2048px is plenty for evidence), JPEG quality around 80. Store the compressed version as the canonical evidence file and hash *that*. Do not keep two versions on device — storage on cheap Android handsets is scarce.

Local storage budget and eviction. Cap the local queue. When it fills, warn loudly rather than silently dropping. A foreman who takes 400 photos over a week without signal must not lose them.

Conflict resolution. Most of this domain is append-only, which spares you the hard cases. Where edits exist (amending a diary entry), use last-write-wins with a full revision history rather than merging — and **never allow silent edits to anything that forms part of a submitted claim.** Once a diary entry is linked to a submitted claim, it should become read-only, with amendments recorded as separate, dated addenda. That is both correct evidentially and much simpler to implement.

Background upload on iOS. iOS is hostile to background work. Use `URLSession` background transfer sessions, expect the OS to schedule them on its own terms, and never promise the user an upload will complete while the app is closed.

Sync protocol sketch

```
POST /v1/sync/push
{
  "device_id": "...",
  "operations": [
    { "op_id": "uuid", "type": "diary_entry.create",
      "payload": {...}, "client_ts": "..." },
    { "op_id": "uuid", "type": "photo.create_metadata",
      "payload": {..., "content_sha256": "...", "size": 1234567} }
  ]
}
-> 200 { "accepted": ["op_id", ...],
         "rejected": [{"op_id": "...", "reason": "..."}],
         "upload_targets": [{"photo_id": "...",
                             "presigned_multipart": {...}}] }

GET /v1/sync/pull?project_id=...&since_cursor=...
-> changes since cursor, cursor-paginated
```

Presigned S3 multipart URLs let the device upload media **directly to object storage**, bypassing your Go backend entirely. That is important for cost and for backend memory. You already run Cloudflare Tunnel as ingress — remember that it was the bottleneck under k6 load testing on TicketHawk. Do not route tens of megabytes of photo through it.

3.5 Media pipeline and storage economics

Photos are the product, which means storage is your dominant variable cost. Model it before you promise anything.

Rough sizing

Assumption	Value
Photos per active site per day	40
Compressed size per photo	400 KB
Working days per month	21
Storage added per site per month	~336 MB
Over a 9-month project	~3 GB

At 200 active sites you are adding roughly 67 GB per month, and you never delete evidence because the whole point is retention. Year three is measured in terabytes.

Controls

Tiered storage lifecycle. Recent evidence in S3 Standard; move to Standard-IA after 90 days; Glacier Instant Retrieval after a project reaches final account. Configure via S3 lifecycle policies.

Thumbnail tiers. Generate a 200px thumbnail and an 800px preview at upload. The app and dashboard should almost never fetch the full-resolution original — only report generation and evidence export should.

Egress discipline. Egress, not storage, is what usually surprises people. Serve thumbnails through a CDN with long cache lifetimes. You already run Cloudflare.

Retention policy in the contract. Decide and publish how long you retain evidence after a subscription lapses. Construction disputes surface years later, so "we delete 30 days after cancellation" is hostile and will lose you customers. A better model: evidence stays readable indefinitely on a free read-only tier; write access requires a subscription. That is a strong retention mechanic and it matches the customer's actual mental model of what they are buying.

Do not say "unlimited." Say "fair use, with typical site volumes included," and define the number.

af-south-1 cost caution. Cape Town region pricing is higher than us-east-1 for most services. Model it against real AWS pricing rather than assuming US figures. The POPIA and latency benefits are worth the premium, but know what the premium is.

3.6 Roles and permissions

Your existing TicketHawk model — `read_only → access → editor → admin → master admin`, flag-based and orthogonal to role — transfers well. Construction needs one extra dimension: **project scoping**, and a distinction between org-internal and external parties.

Role	Scope	Can do
Site agent / foreman	Assigned projects	Create diary entries, photos, compliance records; flag potential claims; cannot submit notices
Contracts manager / QS	Assigned projects	All of the above, plus create and submit claims, generate notices
Org owner	All org projects	Everything, plus billing, party register, org statutory documents
Safety officer / CHSO	Assigned projects, compliance scope	Compliance records, appointments, statutory documents; read-only on claims
Bookkeeper	All projects, financial scope	Variation and claim values, certificates; no safety, no photos
Principal contractor (external)	Their site, across subs	Read subs' compliance status and diary summaries; cannot read subs' claim strategy
Employer / principal agent (external)	Single project, read-only	Receive notices, view shared reports; cannot edit anything

The critical boundary

A principal contractor must never see a subcontractor's draft claims against them. This is the single most important permission rule in the system and it is a trust issue, not a technical one. If a sub suspects the PC can see their claim preparation, they will not use the product.

Make this explicit in the UI. Show the sub exactly what the PC can see, on a dedicated screen. Consider making it a marketing point.

Tenant isolation

Row-level `org_id` on every table, enforced in the data access layer, plus PostgreSQL row-level security as a second line. Every query filtered by tenant. Given your admin console security overhaul experience on TicketHawk — the ungated KYC document routes, the per-capability middleware — apply the same audit discipline here from the start rather than retrofitting.

External party access (PA, employer) should be **scoped share links with expiry**, not accounts. Nobody outside the org wants to create an account to read a report.

3.7 Document generation

Three categories, with different requirements.

Notices and formal correspondence. Legally significant. Must cite the correct clause for the correct contract edition, must be dated, must record delivery. Generate a PDF, email it with delivery and read receipts where possible, store the exact bytes sent, and log the transmission. **The proof of sending is as important as the notice itself.** Store the SES message ID and delivery notification.

Site reports. The daily or weekly PDF sent to the PA or client. This is the visible artefact that makes the product feel valuable. Invest in typography and layout — it is your customer's brand going out to their client.

Claim submission packs. The heavy one: narrative, the substantiating diary extracts, photo appendix, delay calculation, correspondence log. Generate asynchronously via a worker; these can run to hundreds of pages.

Technically: HTML templates rendered to PDF gives you the most control and the fastest iteration. Chromium-based rendering (headless Chrome, or a service) handles complex layout best; `wkhtmltopdf` is lighter but older. Whatever you choose, **the template must be versioned** — a notice generated in 2027 must be reproducible in 2030 exactly as it was sent.

3.8 Security posture

Non-negotiable given what you are storing:

- TLS everywhere; HSTS.
- Encryption at rest on RDS and S3 (SSE-KMS with a customer-managed key, so you can demonstrate key control).
- Evidence bucket: block all public access, no public URLs ever, presigned URLs with short expiry only.
- S3 Object Lock in governance or compliance mode on the evidence bucket — this is what makes "we cannot delete your evidence" a technical statement rather than a promise.
- Separate KMS keys for evidence versus application data.
- Audit log append-only, shipped off-instance.
- Secrets in AWS Secrets Manager or SSM Parameter Store, never in compose files. (You have had a private key exposed once already — treat that as the lesson it was.)
- Rate limiting on auth and on evidence read endpoints.
- POPIA operator agreement in your terms, since you are an operator processing on behalf of contractors who are responsible parties.

PART 4 — STACK AND IMPLEMENTATION

4.1 The stack decision

The original stack document recommends TypeScript, Go, Python, PostgreSQL, SQLite, S3, Redis, a managed event bus, gRPC, OIDC, Docker, GitHub Actions and OpenTelemetry. That is not a bad list — it is a description of the platform you have already built for TicketHawk, which is why it looks right.

The correct move is not to adopt it wholesale. It is to **reuse what you have and delete what you do not need yet**.

Keep

Layer	Choice	Why
Core backend	**Go** (Gin, pgx v5)	You have a production codebase, deployment scripts, and load-test data on this exact stack
Database	**PostgreSQL** (RDS)	Transactional truth, JSONB where the schema needs to flex, strong date arithmetic
Mobile	**Expo / React Native**	Two shipped apps, App Store and Play submissions done, offline sync already solved once
Web	**Next.js + TypeScript**	Four existing front-ends, established patterns
Object storage	**S3** (af-south-1)	Evidence residency for POPIA, Object Lock for immutability
Cache / queue	**Redis**	Already deployed; adequate as a job queue at this scale
Auth	Your existing **JWT + refresh rotation**	Built, hardened, understood
CI/CD	**GitHub Actions + ECR + pull-based deploy.sh**	Working pipeline
Email	**AWS SES**	Migrated already; delivery notifications matter for notice proof
Payments	**Paystack**	Live, payouts approved, handles ZAR and debit orders

Cut, for now

Proposed	Verdict
Python AI services	Phase 4+. PPE detection and progress comparison are research projects. They will consume a year and produce a demo.
Managed event bus	Not until you have measured a bottleneck. PostgreSQL `LISTEN`/`NOTIFY` or a `jobs` table with a Go worker pool covers everything through Phase 3.
gRPC	You have one service. There is nothing to talk to.
Microservice extraction	Modular monolith with clear package boundaries. Extract when you can name the specific service that is scaling differently.
OpenTelemetry	Structured logs plus CloudWatch metrics is enough at this stage. Add tracing when you have a latency problem you cannot explain.
Face recognition / biometrics	Cut permanently. POPIA section 26. Use QR/NFC.
OIDC identity provider	Your existing auth works. Add OIDC later for enterprise SSO when an enterprise asks.

The one genuinely new technical problem is the **notice engine** — deadline arithmetic, calendars, state machines, document generation. That is where your engineering attention should go, and it is pure Go and PostgreSQL.

4.2 What transfers from TicketHawk

Being concrete about reuse, since it materially changes the build estimate:

TicketHawk asset	Reuse in this product
Offline-first SQLite manifest + delta sync (EventsStudio door scanning)	The sync engine core. Different payloads, same architecture. This is the single biggest saving.
JWT auth, bcrypt, refresh token rotation, rate limiting	Drop in near-unchanged
Flag-based staff permission model (`read_only → master admin`)	Extend with project scoping
Rotating HMAC QR ticket system	The same cryptographic instinct applied to evidence hash chains and to QR worker attendance
Organiser KYC / document verification flows	Statutory document upload, verification and expiry tracking
`timefmt.go` SAST handling	Reuse directly; timezone correctness is critical for notices
Push notification system (APNs, FCM)	Deadline alerting
Docker Compose prod deployment + `deploy.sh`	Same pattern
k6 load testing setup	Same methodology
Cloudflare Tunnel ingress (and knowledge of its limits)	Reuse, but route media direct to S3
Analytics event tracking	Product usage instrumentation

Realistically this saves perhaps 30–40% of the build. The domain logic — contract forms, notice rules, claim state machines, compliance registers — is entirely new and is the majority of the work.

4.3 Phase plan

Phase 0 — Domain and validation (weeks 1–12)

No production code. Deliverables:

- Working knowledge of JBCC clause 23 and the Construction Regulations documentation obligations
- Eight to twelve interviews (three CHSOs, five subs, three principal contractors)
- Buyer decision: A, B or C
- A written `notice_rules` table for JBCC main contract and subcontract, reviewed by a QS or attorney
- Sample site diaries collected from three real contractors — these define your schema

Exit criterion: you can describe, without notes, what happens between a contractor noticing a delay and a claim being certified.

Phase 1 — The narrow wedge (weeks 12–24)

One contract form (JBCC). One claim type (EOT). The minimum that proves the thesis.

- Org, users, projects, project parties
- Site diary, two-tier entry, offline-first
- Photos with hash chain, dual timestamps, GPS, watermark
- Claims register with the three-clock state machine
- Notice generation and email delivery with proof of sending
- Deadline alerting with escalation
- Daily/weekly PDF site report
- Confirmation of verbal instruction
- Paystack subscriptions, ZAR, card and debit order

Explicitly **not** in Phase 1: compliance module, multi-contract-form, client portal, AI, analytics, integrations.

Exit criterion: a real contractor tells you the notice clock saved a claim.

Phase 2 — Compliance spine (weeks 24–40)

- Compliance register: toolbox talks, PPE checks, hazard identification, incidents

- Appointments register with expiry
- Statutory document vault with expiry alerting (Letter of Good Standing, CIDB, insurances)
- Regulation 4 notification generation and tracking
- Section 37(2) mandatary agreement register
- QR/NFC worker attendance and induction records
- Safety file assembly and export

Phase 3 — Multi-party and multi-form (weeks 40–60)

- Principal contractor tier: subcontractor accounts, compliance rollup dashboard
- Scoped external share links for principal agent and employer
- GCC and NEC contract forms with their own notice rules
- Variation register and valuation
- Claim submission pack generation
- Rainfall baseline and abnormal-weather flagging

Phase 4 — Depth (year 2+)

- Programme import and critical-path linkage
- Register of Projects / CIDB grading evidence pack
- Accounting integration (Sage, Xero, QuickBooks — check what SA contractors actually use; Sage is more common here than QuickBooks)
- Analytics, benchmarking
- Only now: consider AI for document extraction and search

4.4 Deployment

Continue the TicketHawk pattern:

```
docker compose -f docker-compose.prod.yml up -d --build <service>
```

with a pull-based `deploy.sh` from ECR via GitHub Actions.

Differences to plan for:

- **Media never touches the app server.** Presigned S3 multipart, direct from device.
- **A separate worker container** for PDF generation and deadline recomputation. PDF rendering is memory-hungry and must not compete with API request handling.
- **A scheduled job** running daily to recompute deadlines and dispatch alerts. This is business-critical — if it silently fails, customers miss notices. Alert on job *non-completion*, not just on job failure.
- **Backups with a tested restore.** Evidence in S3 with versioning and Object Lock; RDS automated backups plus periodic logical dumps to a separate account. Test the restore quarterly. Losing a contractor's evidence would end the business.
- Remember the Cloudflare Tunnel scaling lesson from TicketHawk load testing.

4.5 Cost model sketch

Monthly, at roughly 200 active sites:

Item	Rough order
EC2 (app + worker)	t3.medium / t3.large
RDS PostgreSQL	db.t4g.small–medium with storage growth
S3 storage	growing ~70 GB/month, cumulative
S3 requests + egress	driven by thumbnail traffic; CDN-cached
SES	negligible at this volume
Cloudflare	existing
Paystack fees	percentage of revenue

The line that grows without bound is S3 storage, and it grows even from churned customers if you keep evidence readable. Model that deliberately — it is a real long-term liability and also, handled well, a real retention moat.

PART 5 — RESOURCE LIBRARY

Organised by topic. Links were verified at the time of writing. Statutory body entries give the official site; deep links are given where a specific document was consulted.

5.1 South African construction contracts and claims

Primary — read these first

- JBCC (Joint Building Contracts Committee) — the publisher of the Principal Building Agreement. Buy the contract documents. Their advisory notes are free and excellent.
- JBCC Advisory Note on Extension of Time (PDF) — the single most useful free document for understanding the EOT mechanism.
- JBCC Advisory Note 23.0, Edition 6.2 (PDF) — clause 23 in detail, including the adverse weather examples.

Practitioner analysis

- Cox Yeats — JBCC clause 29 extension of time claims — the 20/40 working day mechanism stated plainly.
- Hahn & Hahn — "You mentioned it… but did you notify?" — why notice is fatal.
- Le Roux Consulting — Successful JBCC extension of time claims — the three-step entitlement/quantum/evidence framing.
- Kobus le Roux — 3 steps to successful JBCC EOT claims
- Agora Africa — The principal agent under the JBCC PBA — the principal agent's clocks and the deemed-rejection rule.
- Lexology — JBCC subcontract automatic extensions of time — shorter subcontract notice periods.
- SciELO — Contract administrators' perspective on claim events under the JBCC PBA — peer-reviewed South African research on which claim events actually arise.

Other contract forms

- SAICE (South African Institution of Civil Engineering) — publisher of the GCC.
- Lexology — GCC second edition 2010, weather and EOT — how GCC treats abnormal climatic conditions.
- NEC Contracts — NEC3/NEC4, widely used by South African state-owned enterprises.
- FIDIC — international forms.

Weather and delay

- Plumbing Africa — The influence of weather on construction work and (sub)contractors' obligations — site weather recording obligations, and the SMME compliance gap.
- Long International — Problematic construction contract clauses: weather — international perspective on abnormal weather standards and historical-average benchmarking.
- Construction Management — How do construction contracts deal with rain delays?
- Building in the rain in South Africa — regional rainfall patterns.
- South African Weather Service — climate data source.

Suggested reading

- *Construction Delay Analysis Simplified — The South African Edition*, Dr Hendrik Prinsloo. Referenced in the JBCC advisory note as a step-by-step guide to formulating delay claims. If you read one book on this domain, make it this one.

5.2 Health and safety law and compliance

- Construction Regulations 2014 — full text (PDF)
- ASAQS — Construction Regulations guidelines (PDF) — annotated, including the client/agent appointment duties.
- Regulation 4 — Notification of construction work — the 7-day obligation your users live under.
- Polity — New Construction Regulations 2014 — the original permit thresholds and the SACPCMP approval.
- Labour Guide — Notice regarding application for construction work permit — the exemption notice raising the threshold.
- SAIOSH — construction work permit notice

- CIFA — The Construction Regulations 2014: impact on clients and designers (PDF) — practical walkthrough of the regulation-by-regulation duties.
- Department of Employment and Labour — construction inspection presentation (PDF) — how the inspectorate sees the industry.
- PolicyVault — Construction Regulations 2014
- Labour Guide — general SA labour and OHS resource, useful for tracking change.
- SAIOSH — professional body, publishes regulatory notices.
- Department of Employment and Labour — the regulator.

5.3 COIDA and the Compensation Fund

- CompEasy / CF Online portal — registration and Letter of Good Standing.
- Return of Earnings portal — annual ROE filing.
- ClearComply — COIDA registration and Letter of Good Standing guide 2026 — the annual cycle, deadlines and consequences, clearly set out.
- OurPower — How to register for COIDA — forms, industry classification, penalties.
- Letter of Good Standing — requirements and renewal
- VCA Consulting — COIDA Letter of Good Standing guide

5.4 CIDB and industry registration

- CIDB — the statutory body.
- CIDB — Register of Contractors overview — grading levels, criteria, JV rules, subcontractor position.
- CIDB Construction Registers Service — Register of Contractors, Register of Projects, grading designation calculator, JV calculator, tender opportunities.
- CIDB — Comprehensive guide to contractor registration (PDF)
- CIDB — Grade 2 to 9 application for contractor registration (PDF) — exactly what evidence an upgrade requires.
- CivilSure — Understanding the CIDB rating system — the grade format explained.
- ProTenders — CIDB registration guide 2026
- SACPCMP — registration body for construction health and safety practitioners and project/construction managers.
- ASAQS (Association of South African Quantity Surveyors) — the QS profession; a key influencer group for your product.
- Master Builders South Africa — employer association, regional bodies, member access.

5.5 Data protection and electronic evidence

- Information Regulator of South Africa — POPIA and PAIA regulator.
- POPIA full text, navigable
- POPIA section 26 — prohibition on processing special personal information — the provision that rules out your biometric roadmap items.
- Usercentrics — POPIA overview
- Termly — POPIA explained
- SureDox — sharing documents safely under POPIA — practical redaction and audit-log framing.
- GDPR, BIPA and POPIA compliance for biometric attendance — a vendor piece, but a clear statement of the biometric consent bar.

For ECTA, search for the Electronic Communications and Transactions Act 25 of 2002 on the government gazette or a legislation site and read **section 15** in full. Do not rely on secondary summaries for the provision your entire evidence claim rests on.

5.6 Market and competitor research

- CompanyCam pricing 2026 — complete guide
- CompanyCam 2026 restructure and Beam Finance acquisition

- CompanyCam on G2 · on Capterra · vs JobTread
- Raken review — Contractor ToolStack — pricing ladder, Android quality gap, integration list.
- Raken — Software Advice profile · Capterra · Workyard review
- Raken daily reporting app overview
- Capterra South Africa — contractor management category — what is actually marketed into SA.
- OVS Solutions — Complete guide to workplace safety software in South Africa — the local enterprise landscape and the local-legislation gap.
- GoAudits — construction safety software roundup

Method note: several of the most quotable "the market has a pricing gap" sources are content marketing published by competitors. Read them for facts, not for conclusions.

5.7 Payments and billing

- Paystack documentation — subscriptions, plans, and South African payment methods.
- Stripe global availability — confirms which countries support merchant accounts.
- Stripe alternatives for South African founders — Paddle, Lemon Squeezy, PayFast, Peach Payments compared.
- Using Stripe from South Africa
- Paddle and Lemon Squeezy — merchant-of-record options if you later sell internationally.

5.8 Technical resources

Backend

- Go documentation · Effective Go
- pgx — PostgreSQL driver
- PostgreSQL documentation — particularly date/time functions and row-level security
- PostgreSQL row-level security — second line of tenant isolation
- Redis documentation

Mobile and offline

- Expo documentation
- React Native
- expo-sqlite
- expo-file-system — local media management and storage budgeting
- expo-location — including mock-location detection
- expo-task-manager and expo-background-task — background sync
- SQLite documentation
- Offline First — a community resource on offline-capable application design
- CRDTs explained — useful background even though you probably do not need CRDTs for an append-heavy domain

Storage and media

- Amazon S3 documentation
- S3 multipart upload — resumable uploads
- S3 presigned URLs
- S3 Object Lock — write-once evidence storage
- S3 lifecycle management — tiering to control cost
- AWS pricing calculator — model af-south-1 specifically

Evidence integrity and attestation

- Google Play Integrity API — device and app attestation on Android
- Apple DeviceCheck / App Attest — the iOS equivalent
- RFC 3161 — Time-Stamp Protocol — if you ever need third-party trusted timestamping
- OpenTimestamps — an option for anchoring hash chain roots

Documents and reporting

- Puppeteer — headless Chromium for HTML-to-PDF with full CSS support
- wkhtmltopdf — lighter, older, adequate for simpler layouts
- gopdf and maroto — Go-native PDF generation if you prefer to stay in one language

Infrastructure

- Docker Compose
- GitHub Actions
- AWS SES — particularly event publishing for delivery proof on notices
- Cloudflare Tunnel
- OWASP Application Security Verification Standard — use as your security checklist
- OWASP Top Ten

5.9 Communities and ongoing learning

South African construction

- Master Builders South Africa and its regional associations — the most direct route to contractor members.
- SAIOSH — safety practitioners.
- ASAQS — quantity surveyors, who understand claims better than anyone.
- SACPCMP — construction health and safety practitioners.
- SAICE — civil engineering.
- CIDB — publishes industry reports and statistics useful for market sizing.

SaaS and founder

- Indie Hackers — bootstrapped founder benchmarks.
- r/SaaS and r/Construction
- SaaStr — pricing, retention, go-to-market.

PART 6 — RISKS AND HONEST CAVEATS

6.1 The biggest risk is not technical

You currently operate TicketHawk solo: a live payments platform with production bugs, two apps in store review, an admin console security overhaul in progress and an organiser dashboard retheme underway. TicketHawk has market position and momentum.

Starting a second vertical SaaS is not an engineering decision. It is a decision about where your only genuinely scarce resource — attention — goes. The architecture reuse is real but saves perhaps a third of the build, and the build was never the expensive part. The expensive part is learning an industry, finding customers who are hard to reach, and supporting them while also supporting TicketHawk.

Three honest options:

1. **Do not build it.** Put the same effort into TicketHawk, which already has customers.
2. **Build it slowly, as a validated side track.** Phase 0 only — the interviews and domain learning — costs you evenings, not your business. Decide after twelve weeks with real information.
3. **Build it properly, which means reducing TicketHawk to maintenance.** Only do this if Phase 0 produces a principal contractor who says they will pay.

Option 2 is almost certainly right.

6.2 Specific risks in this plan

Risk	Severity	Mitigation
Deadline engine gives wrong advice and a customer forfeits a claim	Severe — financial harm plus liability	Professional review of the rules; conservative alerting; show workings; never present as authoritative; reviewed terms of service
Contract editions renumber and generated notices cite wrong clauses	High	Version `contract_forms`; never hard-code clause references
Regulatory thresholds change	Medium	Store in data, not code; monitor SAIOSH and DEL
Willingness to pay is lower than modelled	High	Validate in Phase 0 with real price conversations, not surveys
Distribution proves impossible — small contractors unreachable	High	The Buyer B / PC-led model exists precisely to solve this; test it early
A local QS or safety consultant builds the same thing with better domain knowledge	Medium	Move fast on the domain; your engineering advantage only lasts while your domain gap closes
Storage cost grows unbounded	Medium	Lifecycle tiering, thumbnails, defined retention, do not promise unlimited
POPIA exposure processing worker images	Medium	Cut biometrics; operator agreement; af-south-1; minimal retention of faces
CIDB grading angle turns out not to work as hypothesised	Low	Verify against the application forms before building on it

6.3 What this document is not

It is not legal advice, not a specification, and not a substitute for talking to people who have spent careers in this industry. Every regulatory and contractual statement here was checked against a source and linked, but sources go stale, editions change, and secondary commentary is sometimes wrong.

Before any of this becomes code that affects a contractor's money, have it reviewed by:

- a **construction attorney** for the notice regimes and your terms of service
- a **quantity surveyor** for the claim and variation logic
- a **SACPCMP-registered CHSO** for the compliance register
- a **privacy practitioner** for the POPIA position

That review will cost less than one lawsuit and less than one lost customer relationship.

Compiled 1 September 2026. Sources verified at time of writing and linked throughout. Verify at source before relying on any figure, threshold, clause reference or deadline.